Deployment & Customer Handover Guide

Real-Time Deep Learning Road Lane Detection & Autonomous Driving Telemetry System

Framework: MASTER_SDLC_AGENTS | Phase 7 Release | Date: 2026-10-23 | Status: Approved

1. Executive Overview & Autonomous Framework Governance

This deliverable was engineered under the autonomous governance of the MASTER_SDLC_AGENTS 7-Agent Framework. Every phase was strictly validated through Human-in-the-Loop gates, from Product Requirements Document (PRD) to full security audits and containerized release packaging.

2. Deployment Options

Option A: Production Docker Container (Recommended)
• Execute: docker compose up --build -d
• Web URL: http://localhost:8501

Option B: Local Python Virtual Environment
• Setup: python -m venv venv && .\venv\Scripts\activate && pip install -r requirements.txt
• Launch: streamlit run src/app.py

Option C: Cloud Deployment (Streamlit Community Cloud)
• Ready to deploy with packages.txt (libgl1, libglib2.0-0) and requirements.txt.

3. End-User Operational Guide & Telemetry Cards

Dashboard Indicator	State Displayed	Driver Action Required
Camera Tracking	▨ LOCKED (Both Lane Lines Tracked) ◯ SEARCHING (Scanning Road / No Lines Found)	LOCKED: Normal operation. SEARCHING: Pay manual attention; lines obscured.
Vehicle Lane Position	☑ Centered (within 0.12m) ⚠ Drifting Right/Left (> 0.12m)	Steer slightly toward center if drifting badge appears.
Road Shape Ahead	🛣 Straight Road ↪ Gentle Curve ⚠ Sharp Turn	Anticipate steering adjustments based on curve severity.

Driver Safety Alert	✓ SAFE IN LANE (Normal) 🚨 DEPARTURE ALERT! (Flashing red)	Immediate correction required if Departure Alert triggers.